Exploratory Data Analysis

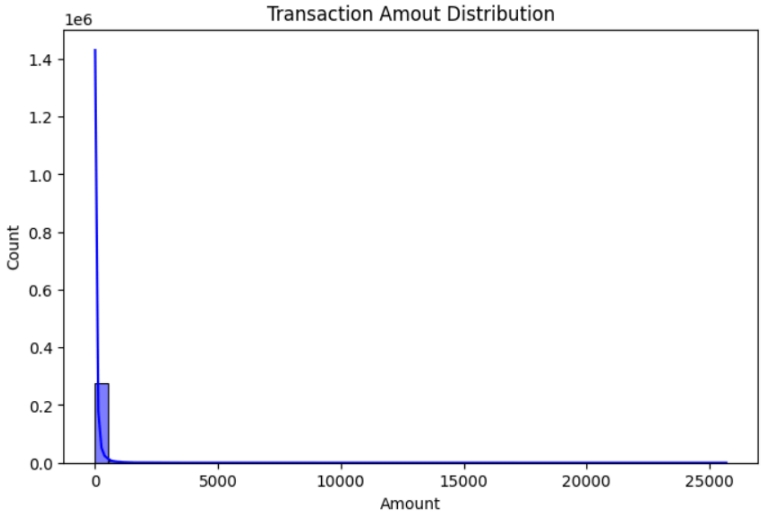
1. Fraud vs Non-Fraud Transactions

Observation: Fraud cases represent less than 0.2% of all transactions.

Business Insight: Fraud detection is a rare-event challenge. Traditional accuracy metrics are misleading because a model could predict “non-fraud” for everything and still appear highly accurate.

Implication: Specialized techniques (e.g., anomaly detection, SMOTE) are required to ensure fraud is not overlooked.

2. Transaction Amount Distribution

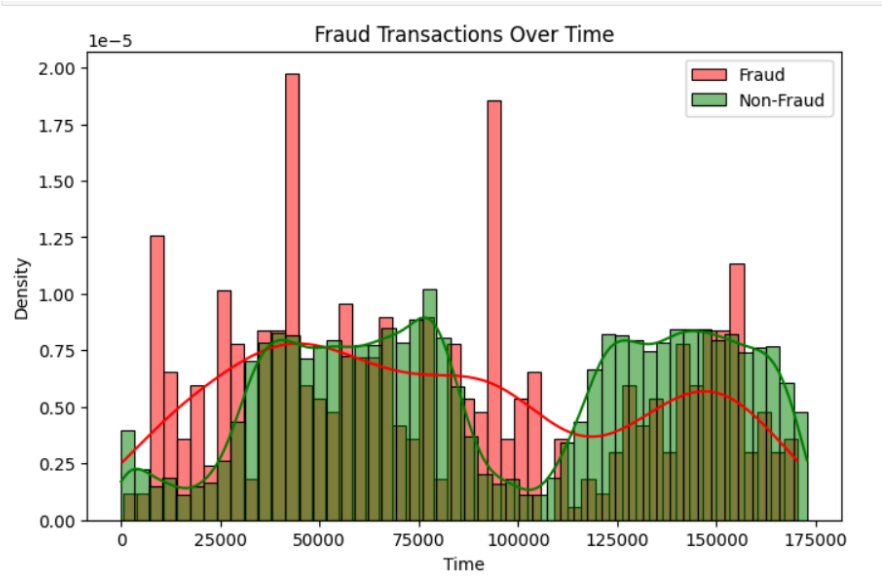


Observation: Most transactions are low-value, while high-value transactions are rare but carry significant financial risk.

Business Insight: Fraud often occurs at extremes — very small charges (testing stolen cards) or very large amounts (maximizing theft).

Implication: Risk-based authentication (OTP, biometrics) should be applied to high-value transactions, while anomaly detection should monitor unusual low-value activity.

3. Fraud Transactions Over Time

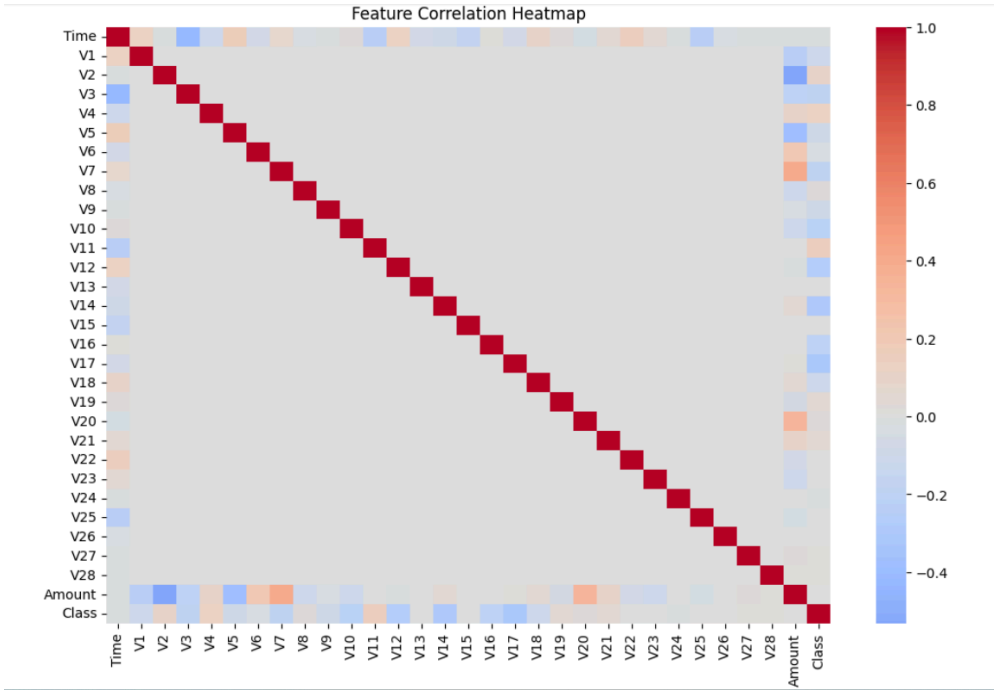


Observation: Non-fraud transactions are evenly distributed, but fraud shows spikes at specific times (e.g., late night, off-peak hours).

Business Insight: Fraudsters exploit unusual time windows when monitoring is weaker.

Implication: Time-based features and real-time monitoring during high-risk periods can strengthen fraud detection.

4. Feature Correlation Heatmap

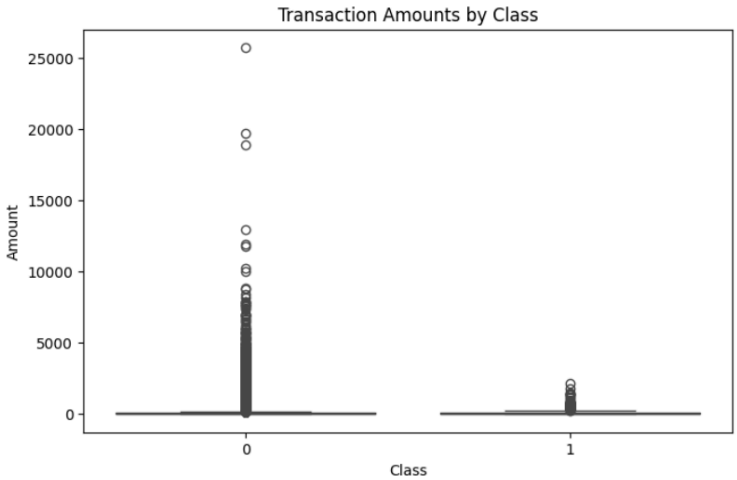


Observation: Certain anonymized features (V1–V28) show strong correlations with fraud.

Business Insight: These features are highly predictive and can guide model training.

Implication: Feature engineering based on correlation analysis improves model efficiency and interpretability, reducing noise in fraud detection systems.

5. Transaction Amounts by Class (Box Plot)



Observation: Non-fraud transactions span a wide range, including many high-value outliers. Fraud transactions are generally lower amounts but still show anomalies.

Business Insight: Distribution differences between fraud and non-fraud provide predictive signals.

Implication: Outlier detection in transaction amounts is a key strategy for flagging suspicious activity.

Conclusion

Fraud detection is not just a technical exercise but a **business-critical priority**. Models must balance **customer convenience** (minimizing false alarms) with **risk management** (catching rare fraud cases). By leveraging transaction distributions, time-based anomalies, and feature correlations, organizations can build robust fraud detection systems that protect customers while maintaining trust and operational efficiency.